



RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

B.Sc. in Applied Sciences

Second Year – Semester II Examination – November / December 2023

CHE 2106 SPECTROSCOPIC METHODS IN ORGANIC CHEMISTRY

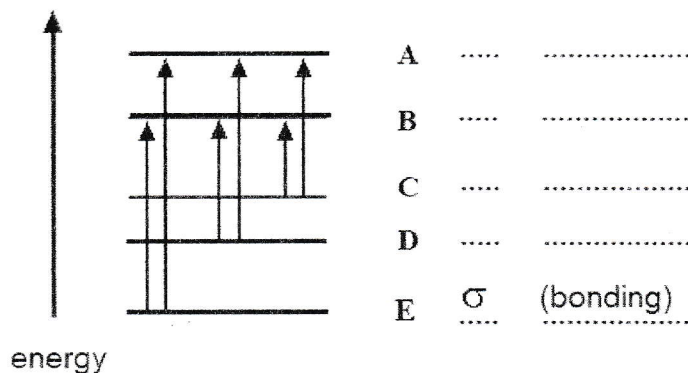
Answer ALL Questions

Time: One (01) hour

1. a) Give a brief account of the relationship between the electromagnetic spectrum and the UV-visible, Infrared, and Nuclear Magnetic Resonance Spectroscopy methods.

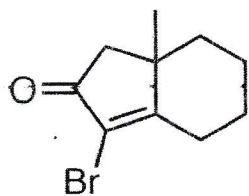
(10 marks)

- b) Some wavelengths of light are absorbed by compounds as they pass through them because the energy of those wavelengths is used to move electrons into higher energy orbitals. The diagram shows the electron transitions which are possible.

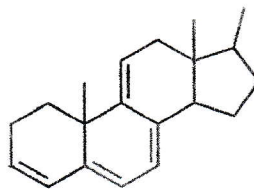


- Give the details of A, B, C and D energy levels. (04 marks)
- Which transition needs the highest energy? (02 marks)
- Which transition absorbs light of the highest frequency? Explain your answer. (03 marks)
- Which transition absorbs light of the highest wavelength? Explain your answer. (03 marks)

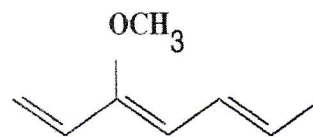
- c) i. Calculate the UV $\lambda_{\max}$ of the following structures using Woodward-Fischer rule. (Basic value for acyclic conjugated enone : 215 nm, heteroannular conjugated system: 214 nm, homoannular conjugated system 253 nm, homodiene component: 39 nm, extended conjugation: 30 nm, Alkyl substitution: 5 nm, Exocyclic double bond: 5 nm, α -substitution: 10 nm, β -substitution: 12 nm)



(i)



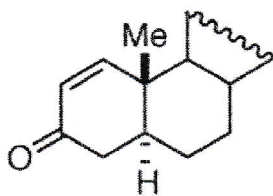
(ii)



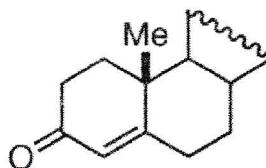
(iii)

(09 marks)

- ii. Distinguish the following compound A from B using UV-visible spectroscopy.



A



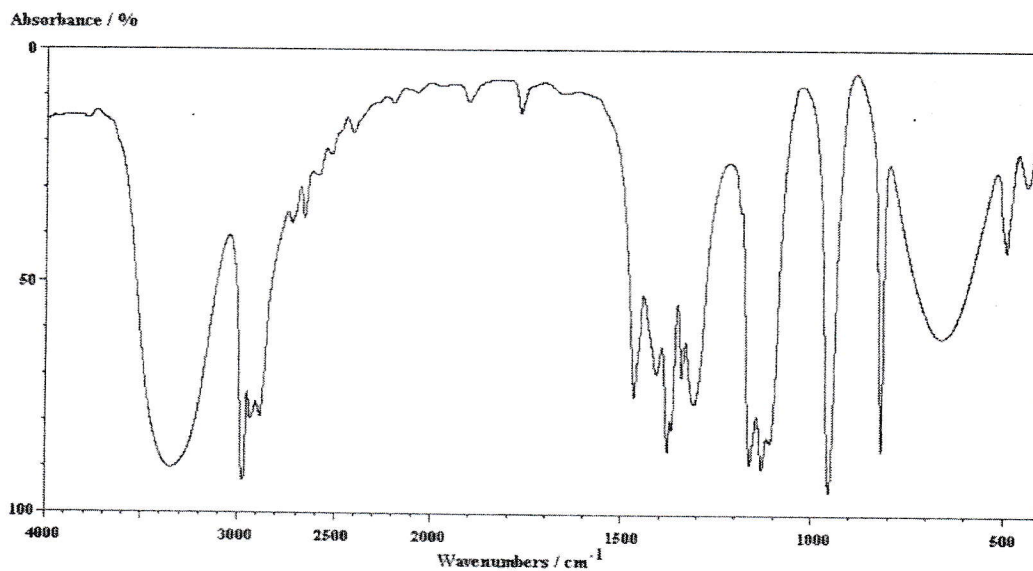
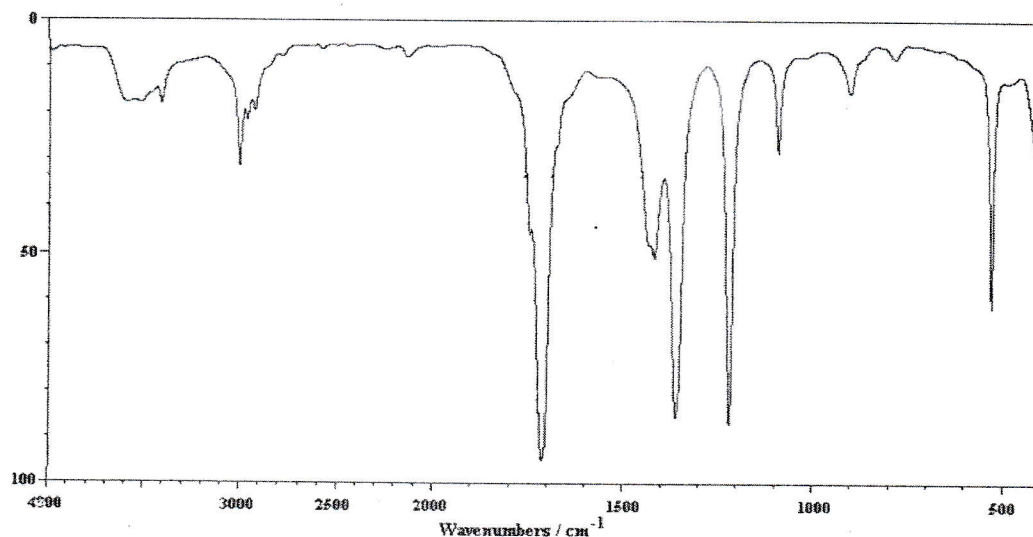
B

(04 marks)

2. a) Write a brief account on "ion and ionization" in mass spectrometry (10 marks)
- b) Discuss the fragmentation of n-pentane and 2-methyl butane, and draw possible mass spectrums for the molecules. (15 marks)
3. a). Asymmetrical stretching / bending and internal rotation change the dipole moment of a molecule, thus IR active. Explain the above sentence. How much movement occurs in the vibration of in a C-C bond in IR? (10 marks)

- b). IR spectra of two compounds with molecular formula C_3H_6O and C_3H_8O are given below. Identify the compounds and relevant spectra, and draw possible structure/s using IR spectra.

(10 marks)



- c). Write a short account and draw H- NMR and ^{13}C - NMR spectrums of methyl acetate

(20 marks)

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